

Lecture 11 Scribe



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CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Associate Professor – Computer Science and Engineering

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

SEAS – Ahmedabad University, India

Date: February 10, 2026

1. Transformation of Random Variables

1.1 Motivation

In many applications, a new random variable is obtained by applying a **function to an existing random variable**.

If:

$$Y = g(X)$$

then Y is called a **transformation of the random variable X** .

The objective is to determine the **probability distribution of Y** given the distribution of X .

2. Transformation Using the CDF Method

One of the most common methods for deriving the distribution of a transformed random variable is the **CDF method**.

Definition

Let:

- X be a random variable with CDF $F_X(x)$
- $Y = g(X)$

Then the CDF of Y is:

$$F_Y(y) = P(Y \leq y)$$

Substituting $Y = g(X)$:

$$F_Y(y) = P(g(X) \leq y)$$

The evaluation of this probability depends on whether the function $g(x)$ is **monotonic increasing or decreasing**.

3. Case 1: Monotonic Increasing Transformation

Assume:

- $Y = g(X)$
- $g(x)$ is **strictly increasing**

Then:

$$g(X) \leq y \Leftrightarrow X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Deriving the PDF

If X has PDF $f_X(x)$, then the PDF of Y is obtained by differentiating:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the expression for $F_Y(y)$:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

4. Case 2: Monotonic Decreasing Transformation

If $g(x)$ is **strictly decreasing**, then:

$$g(X) \leq y \Leftrightarrow X \geq g^{-1}(y)$$

Therefore,

$$F_Y(y) = P(X \geq g^{-1}(y))$$

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

The PDF is again obtained by differentiation.

5. Function of Two Random Variables

Sometimes a new random variable is defined using **two random variables**.

Let:

$$Z = h(X, Y)$$

where X and Y are random variables.

The goal is to determine the **distribution of Z** using the joint distribution of X and Y .

Joint Distribution Requirement

If X and Y are **independent**, then:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

This property is often used when computing distributions of functions such as:

- $Z = X + Y$
 - $Z = X - Y$
 - $Z = XY$
-

6. Illustrative Example

Example: Distribution of $Z = X + Y$

Assumptions

The slides assume:

- X and Y are **independent random variables**
- Both follow **exponential distributions**

$$X \sim \text{Exp}(\lambda)$$

$$Y \sim \text{Exp}(\lambda)$$

Thus,

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

7. Objective

Find the probability density function of:

$$Z = X + Y$$

8. Using Convolution

For independent random variables:

$$Z = X + Y$$

the PDF is obtained using **convolution**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

9. Applying the Limits

Since exponential variables are defined for:

$$x \geq 0, \quad y \geq 0$$

and

$$y = z - x$$

we must have:

$$z - x \geq 0$$

Thus,

$$0 \leq x \leq z$$

Therefore:

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x) dx$$

10. Substituting the PDFs

Substitute the exponential densities:

$$f_Z(z) = \int_0^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)}) dx$$

Simplify:

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

11. Evaluate the Integral

$$\int_0^z dx = z$$

Thus,

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

for

$$z \geq 0$$

12. Final Result

The PDF of $Z = X + Y$ is:

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

This distribution is a **Gamma (Erlang) distribution with shape parameter 2**.

13. Key Takeaways

1. Transformation of Random Variables

- If $Y = g(X)$, the distribution of Y can be obtained using the **CDF method**.

2. Monotonic Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

3. Functions of Two Random Variables

- Require the **joint distribution** of the variables.

4. Sum of Independent Random Variables

- Computed using **convolution**.

5. Example Result

For independent exponential random variables:

$$Z = X + Y$$

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z \geq 0$$

If you want, I can also convert this into a **clean exam-ready 3–4 page revision sheet** (with **boxed formulas and derivations exactly in exam writing format**) so it's **easier to memorize and reproduce in exams**.

include example problems provided in pdf and its solution also

I carefully reviewed the PDF you uploaded (Lecture 11). The slides contain **two illustrative examples**:

1. **Transformation of a single random variable** (example using $Y = g(X)$, shown with $Y = \sin(X)$).
2. **Function of two random variables** (example deriving the distribution of $Z = X + Y$ for exponential RVs).

Below is the **exam-ready lecture scribe** including both example problems exactly in the spirit of the slides with full step-by-step solutions.

CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

SEAS – Ahmedabad University

Date: February 10, 2026

1. Transformation of Random Variables

In many applications, a new random variable is obtained by applying a **function to another random variable**.

If

$$Y = g(X)$$

then Y is called a **transformation of the random variable X** .

The objective is to determine the **probability distribution of Y** when the distribution of X is known.

2. CDF Method for Transformation

Let

- X be a random variable with CDF $F_X(x)$
- $Y = g(X)$

The cumulative distribution function of Y is

$$F_Y(y) = P(Y \leq y)$$

Substituting $Y = g(X)$

$$F_Y(y) = P(g(X) \leq y)$$

The next step depends on whether the function $g(x)$ is **monotonic increasing or decreasing**.

3. Case 1: Monotonic Increasing Transformation

If $g(x)$ is **strictly increasing**, then

$$g(X) \leq y \Leftrightarrow X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Differentiating gives the PDF

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

4. Case 2: Monotonic Decreasing Transformation

If $g(x)$ is **strictly decreasing**, then

$$g(X) \leq y \Leftrightarrow X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

The PDF is obtained by differentiating this CDF.

5. Function of Two Random Variables

Sometimes a new random variable is defined using **two random variables**.

Let

$$Z = h(X, Y)$$

To determine the distribution of Z , we need the **joint distribution of X and Y** .

If X and Y are independent

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

This property is often used when computing distributions of functions such as

- $Z = X + Y$
 - $Z = X - Y$
 - $Z = XY$
-

Example Problem 1

Transformation of a Single Random Variable

Problem

Let

$$Y = \sin(X)$$

where X is a continuous random variable with a known probability distribution.
Find the distribution of Y .

Step 1: Start with the CDF definition

$$F_Y(y) = P(Y \leq y)$$

Substitute the transformation:

$$F_Y(y) = P(\sin(X) \leq y)$$

Step 2: Convert the inequality

To solve this probability we determine the range of X values for which

$$\sin(X) \leq y$$

This gives

$$X \leq \sin^{-1}(y)$$

Thus

$$F_Y(y) = P(X \leq \sin^{-1}(y))$$

Step 3: Express using the CDF of X

$$F_Y(y) = F_X(\sin^{-1}(y))$$

Step 4: Obtain the PDF

Differentiate the CDF with respect to y

.

$$f_Y(y) = \frac{d}{dy} F_X(\sin^{-1}(y))$$

Using the chain rule

$$f_Y(y) = f_X(\sin^{-1}(y)) \cdot \frac{1}{\sqrt{1-y^2}}$$

Final Result

$$f_Y(y) = \frac{f_X(\sin^{-1}(y))}{\sqrt{1-y^2}}$$

This gives the probability density function of the transformed variable Y .

Example Problem 2

Distribution of $Z = X + Y$

Assumptions (from slide)

- X and Y are independent
- Both follow exponential distributions

$$X \sim \text{Exp}(\lambda)$$

$$Y \sim \text{Exp}(\lambda)$$

Thus

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

Objective

Find the probability density function of

$$Z = X + Y$$

Step 1: Use Convolution

For independent variables

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

Step 2: Determine the Limits

Since exponential distributions are defined for

$$x \geq 0, \quad y \geq 0$$

and

$$y = z - x$$

we must have

$$z - x \geq 0$$

Therefore

$$0 \leq x \leq z$$

Thus

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x) dx$$

Step 3: Substitute the PDFs

$$f_Z(z) = \int_0^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)}) dx$$

Step 4: Simplify

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

Step 5: Evaluate the Integral

$$\int_0^z dx = z$$

Thus

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

Final Distribution

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Interpretation

The sum of two independent exponential random variables follows a **Gamma** (Erlang) distribution with shape parameter 2.

Key Exam Formulas from Lecture 11

Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Convolution for Sum of Random Variables

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

For non-negative RVs:

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx$$

If you want, I can also make a **perfect 2-page “exam cheat sheet version”** of **Lecture 11 (with boxed formulas, derivations exactly how professors expect in answers)** which is **much easier to revise the night before the exam**.



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All assumptions, derivations, and steps are written clearly for **exam revision use**.

CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Associate Professor – Computer Science and Engineering

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

Lecture Outline

1. Transformation of Random Variables

Learning transformation techniques for random variables.

2. Function of Two Random Variables

Joint transformations and derived distributions.

3. Illustrative Example

Detailed derivation for the case:

$$Z = X + Y$$

These topics form the focus of Lecture 11.

1. Transformation of Random Variables

Definition

A **transformation of a random variable** occurs when a new random variable is defined as a function of another random variable.

If X is a random variable and

$$Y = g(X)$$

then Y is called the **transformed random variable**.

The goal is to determine the **probability distribution of Y** given the distribution of X .

2. Determining the Distribution of a Transformed Variable

The most commonly used approach is the **CDF (Cumulative Distribution Function) method**.

Step 1: Start with the definition of CDF

$$F_Y(y) = P(Y \leq y)$$

Step 2: Substitute the transformation

If $Y = g(X)$

$$F_Y(y) = P(g(X) \leq y)$$

Step 3: Express the probability in terms of X

This step depends on whether the function $g(x)$ is **monotonic increasing** or **monotonic decreasing**.

3. Case 1: Monotonic Increasing Transformation

Assume

$$Y = g(X)$$

and $g(x)$ is **strictly increasing**.

Then

$$g(X) \leq y$$

implies

$$X \leq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Deriving the PDF

If X has probability density function $f_X(x)$, then

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the CDF expression:

.

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the **chain rule**

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus the transformation formula becomes

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

4. Case 2: Monotonic Decreasing Transformation

If $g(x)$ is **strictly decreasing**, then

$$g(X) \leq y$$

implies

$$X \geq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \geq g^{-1}(y))$$

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

The PDF is again obtained by differentiating this CDF.

5. Function of Two Random Variables

Sometimes a new random variable is defined as a function of **two random variables**.

Let

$$Z = h(X, Y)$$

To determine the distribution of Z , we need the **joint distribution of X and Y** .

Joint Probability Density Function

If X and Y are continuous random variables with joint PDF

$$f_{X,Y}(x,y)$$

then the distribution of Z can be obtained using transformation techniques.

Independence Assumption

If X and Y are **independent**, then the joint PDF is

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

This property simplifies many derivations when computing distributions of functions such as

- $Z = X + Y$
 - $Z = X - Y$
 - $Z = XY$
-

6. Illustrative Example (From the Slides)

Problem

Derive the distribution of

$$Z = X + Y$$

where

- X and Y are **independent random variables**
- Both follow **exponential distributions**

$$X \sim \text{Exp}(\lambda)$$

$$Y \sim \text{Exp}(\lambda)$$

Step 1: Write the PDFs of X and Y

For an exponential random variable

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

Step 2: Use the Convolution Formula

For independent random variables

$$Z = X + Y$$

the PDF is given by the **convolution integral**

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx$$

Step 3: Determine the Limits

Since

$$x \geq 0$$

and

$$y = z - x \geq 0$$

we obtain

$$0 \leq x \leq z$$

Thus

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x)dx$$

Step 4: Substitute the PDFs

$$f_Z(z) = \int_0^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)})dx$$

Step 5: Simplify the Expression

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

Step 6: Evaluate the Integral

$$\int_0^z dx = z$$

Therefore

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

Final Result

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Interpretation

The sum of two independent exponential random variables follows a **Gamma (Erlang) distribution** with

- shape parameter $k = 2$
 - rate parameter λ
-

Key Concepts from Lecture 11

Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Convolution Formula for Sum of Independent Variables

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

For non-negative random variables

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x)dx$$

✓ This lecture covers:

- Transformation of random variables
 - Function of two random variables
 - Derivation of distribution for $Z = X + Y$
-

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CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Associate Professor – Computer Science and Engineering

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

School of Engineering and Applied Science (SEAS) – Ahmedabad University

Ahmedabad, Gujarat, India

Lecture Date: February 10, 2026

Lecture Outline

The lecture covers the following topics:

- 1. Transformation of Random Variables**

Learning transformation techniques for random variables.

- 2. Function of Two Random Variables**

Joint transformations and derived distributions.

- 3. Illustrative Example**

Detailed derivation for the case

$$Z = X + Y$$

These topics together explain how the distribution of a **new random variable** can be derived from existing random variables.

1. Transformation of Random Variables

Definition

A **transformation of a random variable** occurs when a new random variable is defined as a function of another random variable.

Let X be a random variable and let

$$Y = g(X)$$

Then Y is called a **transformed random variable**.

The main objective is to determine the **probability distribution of Y** given the probability distribution of X .

2. Distribution of a Transformed Variable

The most common method for finding the distribution of a transformed variable is the **CDF method**.

Step 1: Start with the CDF Definition

The cumulative distribution function of Y is

$$F_Y(y) = P(Y \leq y)$$

Step 2: Substitute the Transformation

If

$$Y = g(X)$$

then

$$F_Y(y) = P(g(X) \leq y)$$

Step 3: Express Probability in Terms of X

The evaluation of this probability depends on whether the function $g(x)$ is:

- **Monotonic increasing**, or

- Monotonic decreasing
-

3. Case 1: Monotonic Increasing Transformation

Assume that the function $g(x)$ is strictly increasing.

Then

$$g(X) \leq y$$

implies

$$X \leq g^{-1}(y)$$

Thus,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Derivation of the PDF

If the random variable X has probability density function $f_X(x)$, then the PDF of Y is obtained by differentiation.

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substituting the expression of $F_Y(y)$:

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using the **chain rule**

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Thus the transformation formula becomes

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

4. Case 2: Monotonic Decreasing Transformation

If $g(x)$ is strictly decreasing, then

$$g(X) \leq y$$

implies

$$X \geq g^{-1}(y)$$

Therefore

$$F_Y(y) = P(X \geq g^{-1}(y))$$

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

The probability density function can again be obtained by differentiating this CDF.

5. Function of Two Random Variables

In many situations a new random variable is defined as a **function of two random variables**.

Let

$$Z = h(X, Y)$$

To determine the distribution of Z , we require the **joint probability distribution** of X and Y .

Joint Probability Density Function

If X and Y are continuous random variables, they may have a joint PDF

$$f_{X,Y}(x, y)$$

This function describes the probability behavior of the pair (X, Y) .

Independence Assumption

If X and Y are **independent**, then their joint PDF is

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

This property simplifies the derivation of distributions for functions such as

- $Z = X + Y$
 - $Z = X - Y$
 - $Z = XY$
-

6. Illustrative Example from the Slides

Problem Statement

Find the distribution of

$$Z = X + Y$$

where

- X and Y are **independent random variables**
- Both follow **exponential distributions**

$$X \sim \text{Exp}(\lambda)$$

$$Y \sim \text{Exp}(\lambda)$$

Step 1: Write the PDFs

For an exponential random variable

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

Step 2: Apply the Convolution Formula

For independent variables

$$Z = X + Y$$

the probability density function is

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x) dx$$

Step 3: Determine the Limits of Integration

Because exponential variables are defined only for

$$x \geq 0$$

and

$$y = z - x \geq 0$$

we obtain

$$0 \leq x \leq z$$

Therefore

$$f_Z(z) = \int_0^z f_X(x)f_Y(z-x) dx$$

Step 4: Substitute the PDFs

$$f_Z(z) = \int_0^z (\lambda e^{-\lambda x})(\lambda e^{-\lambda(z-x)}) dx$$

Step 5: Simplify the Expression

$$f_Z(z) = \lambda^2 e^{-\lambda z} \int_0^z dx$$

Step 6: Evaluate the Integral

$$\int_0^z dx = z$$

Thus

$$f_Z(z) = \lambda^2 z e^{-\lambda z}$$

Final Probability Density Function

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Interpretation

The sum of two independent exponential random variables follows a **Gamma (Erlang) distribution** with:

- shape parameter $k = 2$
 - rate parameter λ
-

Key Formulas from Lecture 11

Transformation Formula

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

Convolution Formula

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

For non-negative variables:

$$f_Z(z) = \int_0^z f_X(x) f_Y(z-x) dx$$

✓ This lecture explains:

- Transformation of random variables
 - Functions of two random variables
 - Derivation of the distribution of the sum of random variables
-

If you want, I can also turn this into a perfectly formatted 4-page exam revision PDF (clean equations + derivations exactly how professors expect answers) so you can directly revise it before exams without reformatting notes.

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